

ASHISH KUMAR PANDA

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EDUCATION

University of California, San Diego

Master of Science in Computer Science and Engineering - 3.7/4

Sep 2025 – Jun 2027

San Diego, USA

Odisha University of Technology and Research

Bachelor of Technology in Electrical Engineering - 3.98/4

Aug 2019 – May 2023

Bhubaneswar, India

EXPERIENCE

Boolean Lab - UC San Diego Health

AI Research Intern

Sep 2025 – Present

San Diego, CA

- Engineered a **self-supervised PyTorch pipeline** for colon tissue analysis, processing **Spatial Transcriptomics** images to correlate tissue morphology with gene expression concentrations for cancer risk assessment.
- Fine-tuned **DINOv3-7B Vision Transformers** using **LoRA** for feature extraction, reducing memory requirements by **75% (32GB to 8GB)** while maintaining representational competence on limited hardware.
- Integrated a **Mask2Former** segmentation head with trained **LoRA adapters**, achieving an **81% mIoU** and reducing manual annotation time by **90%**, validating the system for automated crypt-level tissue segmentation.

Open Financial Technologies

Software Engineer

Aug 2023 – Aug 2025

Bangalore, India

- Boosted extraction accuracy by **18%** by benchmarking and deploying **Multi modal AI** models via Python, PyTorch, and OpenCV, enabling **92%** of invoices to be auto-converted into e-invoices and cutting manual processing by **75%**.
- Engineered secure fintech solutions using PHP, Laravel, and SQL to develop Payment Pages and automated RBI-compliant risk controls, reducing manual audits by **82%**, cutting fraud by **67%**, and scaling merchant onboarding, while simultaneously implementing RBAC and encryption to protect 50K+ users and guarantee 100% data integrity and uptime during a complex OPEN Payroll HCM vendor migration.

Pucho Digital Healthcare

Deep Learning Research Intern

Mar 2022 – Apr 2022

Remote

- Accelerated mRNA vaccine candidate screening by analyzing antigenicity and cytokine inducibility through **Deep Learning pipelines** implemented in Python, streamlining the selection of high-potential epitopes.
- Improved clinical decision support by implementing **Explainable AI (XAI)** and **OpenCV** to detect cell anomalies in CT/MRI scans, achieving **96.55% sensitivity** while providing interpretable visual heatmaps for medical professionals.

Analytics Labs

Data Science Intern

Apr 2021 – Aug 2021

Remote

- Designed hybrid statistical deep learning models for COVID-19 spread prediction using **Twitter network analytics**, achieving **89% forecast accuracy** and improving early outbreak detection by **23%** over baseline.
- Mitigated financial fraud on imbalanced datasets by engineering detection models using **Genetic Algorithms** and **Scikit-Learn**, attaining **98.5% recall** and **80%+ precision**.

PROJECTS

Retinal Vessel Segmentation for Diabetic Retinopathy | Python, Keras, OpenCV, Google Colab

Apr 2023

- Developed a **Residual U-Net** model for early detection of diabetic retinopathy, applying CLAHE, gamma correction, and other preprocessing methods, and evaluated on DRIVE, STARE, and CHASE_DB1 datasets using binary cross-entropy with Adam optimizer, achieving an **AUC of 0.9667** and **IoU of 0.6334**.

Vocal Bio-Marker | Python, TensorFlow, Librosa, Soundfile, Self-Attention

April 2022

- Created a hybrid **CRNN-Attention** model for non-invasive COVID-19 screening, achieving **97% accuracy** on the ICBHI dataset by utilizing 2D-CNNs for local spectral feature extraction and a **Self-Attention** mechanism to prioritize diagnostic biomarkers over ambient noise.
- Engineered a robust audio processing pipeline using **MFCCs** and their temporal derivatives (deltas) to capture nuanced vocal tract patterns, employing a **Bidirectional LSTM** to model the long-term temporal dependencies of respiratory cycles.

PUBLICATION

- A. K. Panda, S. Dash, A. Kuanar, D. Behera and B. Panda, "Odia Handwritten Numeral Recognition: A Hybrid Modelling Approach," 2021 5th International Conference on Electronics, Communication and Aerospace Technology (ICECA), 2021, pp. 1014-1019, doi: 10.1109/ICECA52323.2021.9675980

SKILLS

Languages & Framework:

Python, R, C++, PHP, Laravel, SQL, TensorFlow/Keras, PyTorch, REST, Hugging Face Transformers, LangChain

Tools & MLOps:

Git, Docker, Kubernetes, Jenkins, MLflow, Sumologic, Sentry, Jira, LaTeX, Linux, Numpy, DINOv3

AI/ML Specialisation:

Large Language Models (LLM), GenAI, RAG, Computer Vision, Vision Transformers (ViT), Multi Modal ML

Relevant Coursework:

Design and Analysis of Algorithms, Operating Systems, Principles of AI : Probabilistic Reasoning and Learning, Recommender System, AI Agents, Computational Bio-informatics